

Part II

Convex Analysis

Lecture Outline

- 1 Convex analysis
- 2 Convex sets
- 3 Convex functions

Convex analysis

In this lecture, we will study two fundamental and very important concepts from convex analysis: convex sets and convex functions. These concepts will be used in later lectures to define and study convex optimization problems, a class of optimization problems which play a major role in optimization theory due to its many appealing properties such as having only global minimizers and being associated with effective computational methods.

Convex sets

Definition

A set $R_c \subset \mathbb{R}^n$ is said to be convex if for every two points $x_1, x_2 \in R_c$

$$\alpha x_1 + (1 - \alpha)x_2 \in R_c$$

for all $0 < \alpha < 1$. A set which is not convex is called non-convex.

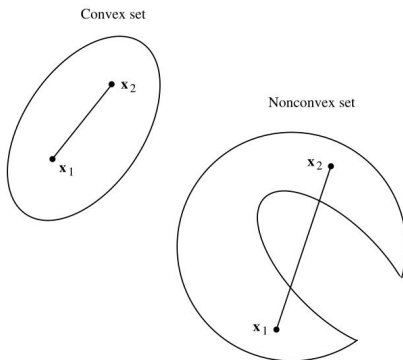


Figure 2.5. Convexity in sets.

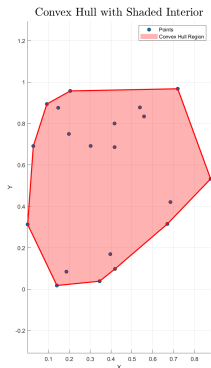
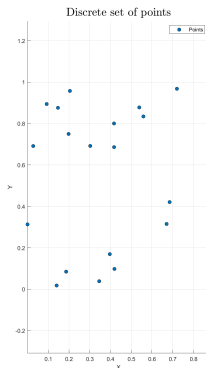
Convex combination, hull and cone

Definition

- ① A convex combination of the points $x_1, \dots, x_m$ in $\mathbb{R}^n$ is a point x on the form

$$x = \sum \alpha_i x_i \quad \text{with } \alpha_i \geq 0 \text{ and } \sum \alpha_i = 1.$$

- ② The convex hull of a subset $R \subset \mathbb{R}^n$ is the (convex) set of all convex combinations of points in R .



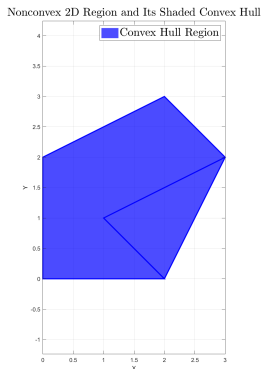
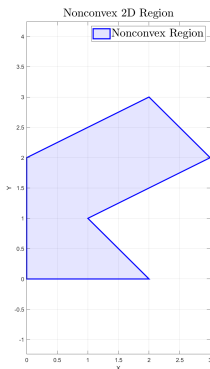
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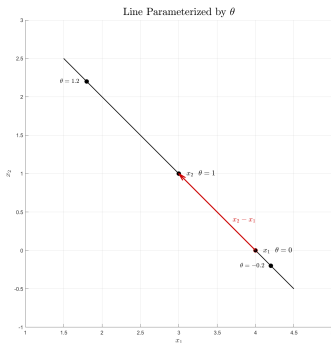
Example

Vector (sub)spaces

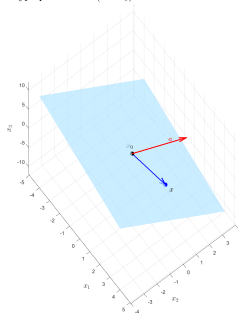
- $\mathbb{R}^n$
- $\text{Range}(A) = \{z \in \mathbb{R}^n \mid z = Ax, A \in \mathbb{R}^{m \times n}\}$
- $\text{Null}(A) = \{x \in \mathbb{R}^n \mid 0 = Ax, A \in \mathbb{R}^{m \times n}\}$

Affine subspaces

- Line (through $x_1, x_2 \in \mathbb{R}^n$): $\{z \in \mathbb{R}^n \mid z = \theta x_1 + (1 - \theta)x_2, \theta \in \mathbb{R}\}$
- Hyperplane: $\{x \in \mathbb{R}^n \mid a^\top x = b, a \in \mathbb{R}^n, b \in \mathbb{R}\}$



Hyperplane $0 = a^\top(x - x_0)$ with normal vector a



Example

- A (closed) half-space:

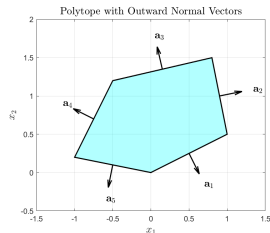
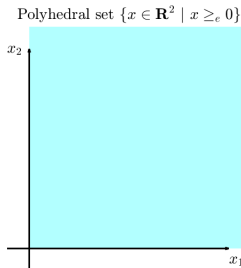
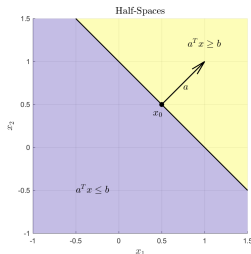
$$\{x \in \mathbb{R}^n \mid a^\top x \leq b, a \in \mathbb{R}^n, b \in \mathbb{R}\}$$

- A (convex) polyhedral set: $\bigcap_{i=1}^k H_i$ with H_i a half-space:

$$\{x \in \mathbb{R}^n \mid Ax \leq_e b, A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m\}$$

with $\leq_e$ denoting component-wise

- A (convex) polytope: a bounded (convex) polyhedral set.

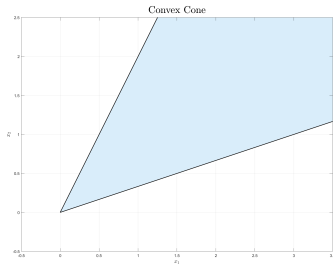


Example

Convex cone: A subset $C \subset \mathbb{R}^n$ is a convex cone if

$$\alpha x + \beta y \in C,$$

for all $\alpha, \beta > 0$ and $x, y \in C$.



Example

Operations preserving convexity

- Intersections: If R_{c1} and R_{c2} are convex sets then $R_{c1} \cap R_{c2}$ is a convex set
- Affine functions: If R_c is a convex set and $f(x) = Ax + b$ then $f(R_c)$ is a convex set

Definition

A function

$$f : \text{dom}(f) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}; \quad x \mapsto f(x)$$

is said to be convex on $R_c \subseteq \text{dom}(f)$ if

- R_c is convex, and
- for every two points $x_1, x_2 \in R_c$

$$f(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha f(x_1) + (1 - \alpha)f(x_2),$$

for all $0 < \alpha < 1$.

If moreover

$$f(\alpha x_1 + (1 - \alpha)x_2) < \alpha f(x_1) + (1 - \alpha)f(x_2), \quad x_1 \neq x_2,$$

for all $0 < \alpha < 1$, the f is said to be strictly convex.

A (strictly) concave function f is one with the property that $-f$ is (strictly) convex.

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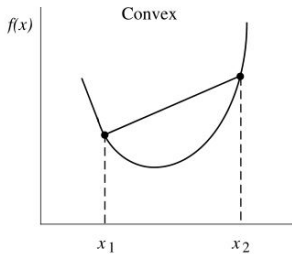
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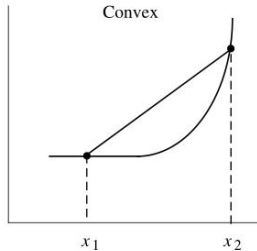
Technical remark: A function convex on R_c is continuous on $\text{int}(R_c)$, however, it may be discontinuous on the boundary as demonstrated by the following example: Let $R_c = [0, 1]$, and $f(x) = 0$ for $x \in \text{int}(R_c) = (0, 1)$ and $f(\{0, 1\}) = 1$.

Example

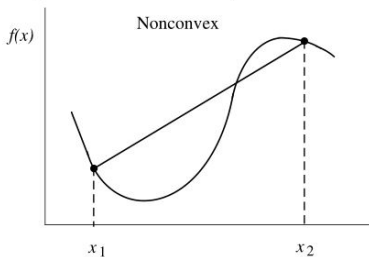
Convex functions



(a)



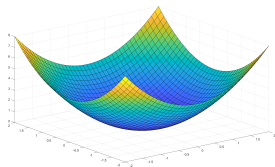
(b)



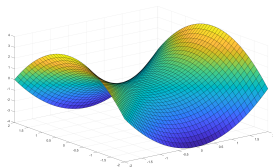
(c)

Convex functions

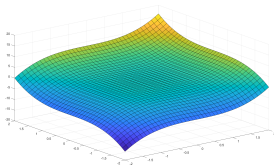
Example



(j) Convex



(k) Nonconvex



(l) Nonconvex

Convex functions

Theorem

- A function f is convex on R_c iff its epigraph

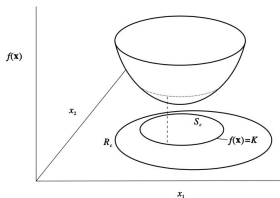
$$\text{epi } f = \{(x, \mu) \mid \mu \geq f(x)\}$$

is a convex set.

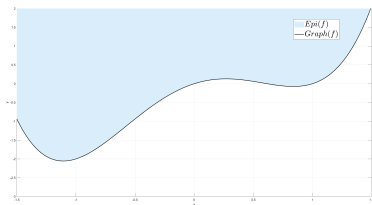
- If f is a (strictly) convex function on R_c and $K \in \mathbb{R}$, then

$$\{x \mid f(x) \leq K\} \subset R_c, \quad (\{x \mid f(x) < K\} \subset R_c),$$

is a convex set.



(m) Convex preimage of f



(n) Epigraph of f

Theorem

- If f_1 and f_2 are convex functions on R_c and $a, b \geq 0$, then

$$f = af_1 + bf_2$$

is convex on R_c .

- If f_1 and f_2 are convex functions on R_{c1} and R_{c2} respectively, then

$$f(x) = \max\{f_1(x), f_2(x)\}$$

is convex (on $R_{c1} \cap R_{c2}$).

- If $f : R_c \times Y \rightarrow \mathbb{R}$ and $x \mapsto f(x, y)$ is a convex functions for each $y \in Y$, then

$$g(x) = \sup_{y \in Y} \{f(x, y)\}$$

is convex (on R_c).

- If $h : \mathbb{R}^k \rightarrow \mathbb{R}$ is convex and nondecreasing (resp. nonincreasing) in each argument, and $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $i = 1, \dots, k$ is convex (resp. concave), then

$$f(x) = h(g(x))$$

is convex with $g(x) = (g_1(x), \dots, g_k(x))$.

Convex functions

In the sequel, we write $M > 0$ (resp. $M \geq 0$) to indicate that the symmetric $n \times n$ matrix $M \in \mathbb{R}^{n \times n}$ is positive definite (resp. positive semi-definite).

Theorem

- Let $f \in C^1$. Then f is a (strictly) convex function on R_c iff for all $x, x' \in R_c$

$$\begin{aligned} f(x) &\geq f(x') + \nabla f(x')^\top (x - x'), \\ (f(x) &> f(x') + \nabla f(x')^\top (x - x')). \end{aligned}$$

- Let $f \in C^2$. Then f is a (strictly) convex function on R_c iff $H_f(x) \geq 0$ ($H_f(x) > 0$) for all $x \in R_c$.

Example

- $f(x) = \sin(x)$ is convex on $R_c = [-\pi, 0]$
- $f(x) = \sin(x)$ is strictly convex on $R_c = (-\pi, 0)$
- $f(x) = \sin(x)$ is nonconvex on $R_c = [-\pi, 0.5]$

Common convex functions in machine learning

Exponential Function

- **Form:** $f : \mathbb{R} \rightarrow \mathbb{R}; x \mapsto f(x) = e^x$
- **Convexity:** Convex because its second derivative (e^x) is always positive.
- **Applications:** Softmax function for probabilities, exponential family distributions, feature transformation in boosting algorithms

Negative Logarithm

- **Form:** $f : \mathbb{R}_{>} \rightarrow \mathbb{R}; x \mapsto f(x) = -\log(x)$
- **Convexity:** Convex because its second derivative ($\frac{1}{x^2}$) is positive.
- **Applications:** Log-loss (negative log-likelihood) in logistic regression, Kullback-Leibler (KL) divergence in probabilistic models

Common convex functions in machine learning

Affine Linear Functions

- **Form:** $f : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x) = w^\top x + b$
- **Convexity:** Convex because their Hessian is zero.
- **Applications:** Linear regression, decision boundaries in support vector machines (SVMs), intermediate transformations in neural networks

Quadratic Functions

- **Form:** $f : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x) = \frac{1}{2}x^\top Qx + b^\top x + c$
- **Convexity:** Convex if $H_f(x) = Q \geq 0$.
- **Applications:** Ridge regression (L2 regularization), loss functions in support vector machines

Common convex functions in machine learning

Regularization Functions

L1 Norm (Lasso Regularization)

- **Form:** $f : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x) = \|x\|_1 = \sum_{i=1}^n |x_i|$
- **Convexity:** Convex because it is a sum of absolute values.
- **Applications:** Sparsity in model weights, feature selection

L2 Norm (Ridge Regularization)

- **Form:** $f : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x) = \|x\|_2^2 = x^\top x = \sum_{i=1}^n x_i^2$
- **Convexity:** Convex because it is a quadratic function with $H_f(x) = I > 0$.
- **Applications:** Preventing overfitting, weight decay in neural networks

More generally, all the following norms are convex functions

- $\|\cdot\|_p : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto \|x\|_p = (\sum_i |x_i|^p)^{1/p}, p=1,2,\dots$
- $\|\cdot\|_\infty : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto \|x\|_\infty = \max_i |x_i|$

Common convex functions in machine learning

Elastic Net Regularization

- **Form:** $f : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x) = \alpha \|x\|_1 + (1 - \alpha) \|x\|_2^2, 0 \leq \alpha \leq 1$
- **Convexity:** Convex because it is a weighted combination of two convex functions.
- **Applications:** Hybrid feature selection and regularization

Log-Sum-Exp Function

- **Form:** $f : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x) = \log \left(\sum_{i=1}^n e^{x_i} \right)$
- **Convexity:** Convex because its Hessian is positive semi-definite.
- **Applications:** Softmax function in neural networks, Margin-based probabilistic models

Common convex functions in machine learning

Hinge Loss

- **Form:** $f : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x) = \max(0, 1 - y(w^\top x + b))$,
- **Convexity:** Convex as it is a maximum of affine functions.
- **Applications:** Used in support vector machines for classification

ReLU (Rectified Linear Unit)

- **Form:** $f : \mathbb{R} \rightarrow \mathbb{R}; x \mapsto f(x) = \max(0, x)$,
- **Convexity:** Convex as it is a maximum of affine functions.
- **Applications:** Activation functions used in neural networks to introduce non-linearity (helps mitigate the vanishing gradient problem).

Common convex functions in machine learning

Absolute Error (L1 Loss)

- **Form:** $f : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x) = \sum_{i=1}^n |y_i - x_i|$
- **Convexity:** Convex as it is the sum of absolute values.
- **Applications:** Robust regression, sparsity-inducing models (e.g., Lasso)

Mean Absolute Error (MAE)

- **Form:** $f : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x) = \frac{1}{n} \sum_{i=1}^n |y_i - x_i|$
- **Convexity:** Convex as it is the sum of absolute values.
- **Applications:** Regression tasks, loss function for continuous outputs

Mean Squared Error (MSE)

- **Form:** $f : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x) = \frac{1}{n} \sum_{i=1}^n (y_i - x_i)^2$
- **Convexity:** Convex because it is a quadratic function with $H_f(x) = \frac{1}{n} I > 0$.
- **Applications:** Regression tasks, loss function for continuous outputs

Common convex functions in machine learning

Cross-Entropy Loss

- **Form:** $f : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x) = -\sum_{i=1}^n y_i \log(p_i(x))$, where
 - $p_i(x) = \frac{e^{x_i}}{\sum_j e^{x_j}}$ is the i th coordinate function of the softmax function
- **Convexity:** Convex for binary $y \in \{0, 1\}^n$ or multi-class $y \in \mathbb{N}^n$ classification problems.
- **Applications:** Classification tasks in logistic regression and neural networks

Negative Log-Likelihood

- **Form:** $f : \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x) = -\sum_{i=1}^n \log(p(y_i|x_i))$
- **Convexity:** Convex for appropriate probabilistic models p .
- **Applications:** Logistic regression, probabilistic graphical models

Common Nonconvex functions in machine learning

- **(logistic) Sigmoid Function:** $f : \mathbb{R} \rightarrow \mathbb{R}; x \mapsto f(x) = \frac{1}{1+e^{-x}}$
- **Applications:** Logistic regression for binary classification, neural networks as an activation function (though less common now due to vanishing gradient issues).
- **Swich Function:** $f : \mathbb{R} \rightarrow \mathbb{R}; x \mapsto f(x) = \frac{x}{1+e^{-x}}$
- **Applications:** Modern deep neural networks for tasks like object recognition. Smooth activation function that improves convergence.
- **Hyperbolic Tangent Function:** $f : \mathbb{R} \rightarrow \mathbb{R}; x \mapsto f(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$
- **Applications:** Neural networks as an activation function (used in hidden layers).